Name .....Centre/Index No...../.....

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P525/1
CHEMISTRY
Paper 1
August, 2026
$2\frac{3}{4}$ hours.

JINJA JOINT EXAMINATIONS BOARD

Uganda Advanced Certificate of Education

MOCK EXAMINATIONS –AUGUST, 2026

CHEMISTRY

(Principal Subject)

Paper 1

2 hours 45 minutes.

INSTRUCTIONS TO CANDIDATES:

- This paper consists of **TWO** Sections A and B.
- Section A has **TWO Compulsory** Items.
- Section B has two parts, I and II.
- Respond to only **ONE** Item from each part.
- Responses to Section A and B must be written
- in the answer booklet(s) provided.
- Respond to **FOUR** Items in all.

BEGIN EACH QUESTION ON A FRESH PAGE

No	Basic Code	Weighted Score	Initials
1			
2			
3/4			
5/6			

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Item 1

Electrochemical cells generate energy from spontaneous redox reactions. Their importance lies in driving modern technology through energy storage, industrial manufacturing etc. A group of S.5 students under their science club have undertaken a project to select the strongest oxidizing and reducing agents between Copper and Zinc, construct an electrochemical cell and apply quantitative chemical principles to make recommendations but they could not proceed. When they consulted their chemistry teacher, he provided them with the following thermochemical and electrochemical data

First ionization energy of Zn	906kJ mol^{-1}
Second ionization energy of Zn	1733.3kJmol^{-1}
Hydration energy of $Zn^{2+}(aq)$	2046kJmol^{-1} First
ionization energy of Cu	754kJmol^{-1}
Second ionization energy of Cu	1958kJmol^{-1}
Hydration energy of $Cu^{2+}(aq)$	2100kJmol^{-1}

Half cell	**E°/V**
$Zn(s) \mid Zn^{2+}(aq)$	+0.76
$Cu(s) \mid Cu^{2+}(aq)$	-0.34

Task

The S.5 students are now challenged and have approached you. ***Task***: As a chemistry student analyse and interpret the data to address the S.5 issues.

Item 2

A manufacturing company in Uganda uses compound X as a primary intermediate for making a dye which gives blue jeans their iconic colour. The laboratory attendant for the company after her analysis discovered that X is immiscible with water, contains 77.3% C, 7.58% H and the rest nitrogen and that a mixture of X (boiling point 180°C) and water (boiling point 100°C) when steam distilled at 760 mmHg and 98°C gave a distillate which contained 85% by mass water.

Management of the company is now interested in molar mass of X, molecular formula of X and the associated applications of steam distillation. (Hint: Saturated vapour pressure of water at 98°C is 734.9mmHg).

The laboratory attendant is now challenged. Your teacher came across this information.

Task:

As a student of chemistry analyse and interpret the available data to address management concerns.

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SECTION B PART I

*(Attempt only **one** Item from this part)*

Item 3

The chemistry of period 3 elements is vital to the foundation of modern technology, biology and industry. A telecom engineer advised that how periodic trends affect reactivity and reactions of the elements with water, alkalis and dilute acids is very important in modern technology.

A manufacturing plant dealing in modern technology equipment is interested in how periodic properties of elements affect reactivity, reactions of elements and the health implications of any of the elements. An industrial chemist provided the plant management with the following data:

Element	Na	Mg	Al	Si	P	S	Cl
Electron affinity ($kJmol^{-1}$)	-20	+67	-30	-135	-60	-200	-364
Ionization Energy ($kJmol^{-1}$)	492	743	579	791	1060	1003	1254

Your laboratory technician came across this information but he is challenged

Task:

As a student of Chemistry prepare a writeup to address the management's concerns

Item 4

The chemistry of Group 17 elements is important for water sanitation, health and industrial manufacturing. An environmental chemist advised that how periodic properties affect reactivity of the elements, reactions of elements with water and alkalis play a great role in their applications in daily life and provided the following data.

Element	F_2	Cl_2	Br_2	I_2
Atomic radius (nm)	0.072	0.09	0.114	0.133
Boiling point (^0C)	-188	-35	59	184

A manufacturing company dealing in antiseptics is interested in how periodic trends affect reactivity, reactions of the elements and health implications if any of any element. Your teacher came across this information.

Task:

As a Chemistry student, analyse and interpret the available data to address the company's issues.

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PART II

*(Attempt only **one** Item from this part)*

Item 5

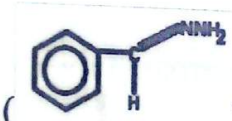
Use of nail polish removers and polymers is increasing in Uganda. A chemical plant makes polythene and ethyl acetate, a nail polish remover from propene and methane respectively. As a way of mitigation, excess raw propene is reacted with bromine water while excess methane is reacted with chlorine gas in UV light.
The plant Management is interested in the suitability of its products for their applications, analysis of its mitigation reactions with equations, an evaluation of its synthetic pathways and the possible health implications of its products. Your chemistry teacher came across this information.

Task:

As a student of chemistry make a write up to address the plant management concerns.

Item 6

A school science club wants to produce benzaldehyde hydrazone



( ) from ethane and propanone (solvent) from propan-1-ol.
One student raised the issue of the managing excess propan-1-ol and the product propanone. The chairperson proposed reacting propanone with 2,4-dinitrophenylhydrazine and propan-1-ol with hot concentrated sulphuric acid as a mitigation measure.
The report was subjected to the patron of the science club who is interested in the suitability of propan-1-ol for its application, analysis of mitigation reaction with equations, an evaluation of synthetic pathways with associated health implications.

Task:

As a student of chemistry analyse and interpret the data to address the club's issues.

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